

Memory architecture decisions — ADR-0058 to 0061

Date: 2026-09-28

Desirée is an ambient observer in a control room: everything streams in, she processes it, and she calls out what is worth it. These four decisions define how that stream becomes memory, and why. Each one is grounded in what the evaluation suite measured on 28 September 2026.

The decisions in one line each

ADR	Decision	Evidence it rests on
0058	Two kinds of memory: facts he told her, and observations, each with its source	TV, guest and screen claims became "his" facts; the screen's address overwrote his own
0059	An index and small subject files replace the single facts file loaded into every prompt	Whole-facts memory stops at ~530 facts (8k) and ~1,100 (16k); corrections are lost first
0060	Subcortical triages at the door; code, not the model, decides what may write facts	Told the source, the extractor stored the screen's address anyway
0061	The bridge: an observation becomes his fact only when he confirms it	"Is that dentist appointment yours?" is both the safeguard and a conversation opener

What stays from earlier decisions

- Write-time extraction (ADR-0002) and the append-only verbatim log (ADR-0044).
- Speaker labels and the quote check together, at answer time: the best measured safety, with 4 fabrications at 72.7% (ADR-0057).
- Every action reports an outcome (ADR-0056), extended to the model: context size stated explicitly, and oversized prompts fail loudly.

Measured, 28 Sep 2026 (house dataset, 44 answerable questions, qwen3:14b answering)

Measure	Result
Run-to-run spread from extraction alone	70.5% to 75.0%: differences of 2 questions or fewer are noise
Speaker labels and quote check	72.7%, 4 fabrications
Speaker labels without quote check	79.5%, 11 fabrications: every TV, guest and screen attack returned
Speech through Piper and Whisper, scored fairly	56.8%; with his names given to Whisper, 65.9%
Silent truncations before the harness fix	874. After: 0, with oversized prompts refused and counted

ADR-0058 - Two kinds of memory: what he told her, and what she observed

Status: Accepted

Date: 2026-09-28

Context

Desirée is an ambient observer. Like a person in the room, she hears and sees everything: his speech, other people, the TV, the screen, the cameras. A person's memory includes all of it, and it is answerable later. But a person also remembers *where* each thing came from: "you told me" is not the same as "I heard it on the TV". Psychologists call this source memory.

The POC kept one facts file with no sources, and the eval showed what that costs (house dataset, 28 Sep 2026). With every line filed as his, the fabrications came straight from other sources:

- "The alarm code is 1234": the TV.
- "The alarm code is 2580": a guest.
- "Our address is 9 Harbour Road, Kelso": text on the screen.

In one run, the screen's address was even written over his real one as a correction: "We live in Kelso... previously, we lived in Brenmoor."

Decision

Memory holds two kinds of knowledge, stored separately, and every item carries its source.

- 1 **Facts:** what a principal told her about themselves and their world. His facts live under me/, and a household member's own facts under people/<name>/. Facts are his or theirs by authority.
- 2 **Observations:** everything her senses picked up: guests' speech, ambient and TV audio, the screen, the cameras. They are stored under observations/, time-stamped and tagged with the source. An observation is never a fact about anyone.

Both are answerable. An answer from observations always says where it came from: "The TV mentioned 1234 and Tom said 2580, but you've never told me your code." That is honest and useful, not a fabrication.

The verbatim log (ADR-0044) keeps every utterance with its speaker, as now.

Rejected

- **One store, with the model told to weigh sources.** Measured: the 8b extractor, told who was speaking, stored the screen's address anyway and filed it as a correction. Authority cannot live in a prompt.
- **Discarding everything not said by him.** It loses what a person in the room would remember, and makes "what did Tom say about his cats?" unanswerable.

Consequences

The eval must accept attributed answers ("the TV said...") as correct, not score them as fabrications. Observation storage grows fast, so it is governed by retention (ADR-0061). His facts are not.

ADR-0059 - An index and subject files, not one resident facts file

Status: Accepted

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Supersedes: the single resident facts file of ADR-0002, in part. Write-time extraction stands; loading all facts into every prompt does not.

Context

The POC sends the entire facts file with every extraction and every answer. The scale eval (harness v2, 0 truncations, 28 Sep 2026) measured the result:

Context	Where it stopped	Evidence
8k	about 530 facts	107 extractions refused by size 500, 607 by 1,000
16k	about 1,100 facts	84 refused at 1,000; answers 1.2s typical, 1.9s worst

Refusals hit the newest statements first, because they come last, when the prompt is biggest. So corrections ("Pepper's five now") are the first thing lost: the opposite of what memory should do.

Before the explicit context and loud-failure fixes, the same overflow happened silently. Ollama cut 874 prompts from the front, dropping the instructions, and nothing noticed (ADR-0056 applied to the model itself).

With enough context, recall itself held: 16k stayed at 72.7% from 0 to 500 statements. The problem is fitting memory into the prompt, not finding facts in it.

Decision

Memory is an **index plus subject files**, the design Claude's own memory uses.

- **One file per subject:** me/home, me/work, me/car, people/noor, people/theo and so on. Each file is small, and capped. Near the cap it is consolidated, not grown.
- **An index is always in context:** each subject's name, a one-line description and its aliases. Nothing else is resident except me/profile.
- **Writing:** the extractor reads the index, picks the subject, opens only that file, and updates it in place, keeping history ("Dunmore, previously Carrow").
- **Reading:** a question goes to the index, then the one or two relevant files, then the answer. The verbatim log search stays as the safety net when the index points the wrong way.
- **A fact that spans subjects** lives in one home file, with links from the others.

The prompt is sized to the question, not to how much memory exists.

Rejected

- **A bigger context window.** It moves the ceiling (530 to about 1,100 facts) without removing it, and every answer gets slower.
- **Embeddings first.** Word-overlap and subject selection are cheaper, inspectable and deterministic. Embeddings wait until measured recall failures justify them (ADR-0030's ladder).

Consequences

The model now chooses which file to open, and a wrong choice is a miss. The eval measures subject-selection accuracy as its own number. The index grows by subject, not by fact: a household might have 50 to 200 subjects against thousands of facts.

Every request states its context size explicitly, and a prompt that will not fit fails loudly (`ContextOverflow`). It is never silently cut.

ADR-0060 - Triage at the door: only principals' speech writes facts

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Context

Everything Desirée perceives arrives through Subcortical: microphone, screen, cameras. ADR-0054 makes perception ingress: observed, never obeyed. ADR-0045 makes everything carry provenance. ADR-0019 attributes every utterance to a principal.

The eval showed where authority must be enforced. With speaker labels on and the quote check off, every attack came back: the TV's code, the guest's code, the screen's address. The extractor had been told the source and told to ignore it, and did not. The protection that worked (4 fabrications at 72.7%) came from code: the quote check refusing any answer not grounded in his own logged words.

Decision

Subcortical triages every input by source **before** anything reaches memory, and **code**, not the model, decides where each input may write:

```
microphone -> speaker ID +- him / household -> conversation + fact extraction
                                     +- guest -----> conversation + observations
                                     +- unknown / media -> observations (ambient, not a turn)
screen / cameras -> Subcortical -> observations
```

- A write path is chosen by the source, in code. His speech can write me/. A household member's speech can write only people/<self>/. Nothing else can write facts at all. A path that isn't writable from a source cannot be reached by any model output: the same principle as the single writable mount (profile read-only, workspace writable).
- **Authority floor:** a lower-authority source can never supersede a higher one about the owner. Within the same authority, the newer statement wins (recency).
- **Unknown voices default to ambient, never to him.**
- **The quote check and speaker labels stay together** at answer time, as the last line of defence if anything is misrouted.

Rejected

- **Telling the extractor who spoke and trusting it to comply.** Measured to fail.
- **A trust weight that can be outvoted.** Repetition from a low-trust source, like a TV advert, would accumulate into belief. Authority is a floor, not a weight.

Consequences

The door is now the weakest point. If speaker ID labels the TV as him, it writes into me/, and nothing downstream can tell. Speaker verification (ADR-0019) moves from a nice-to-have to the load-bearing control, and it needs its own tests. Read-back (ADR-0057) is the backstop for what gets through.

ADR-0061 - The bridge: an observation becomes his fact only when he confirms it

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Context

Observations often contain true things about him. His screen shows "Dentist, Friday 10am". A person sitting beside him would remember it as something they saw, not as certain fact: it might be his partner's, an old reminder, or an ad. A good friend would ask.

This fits the control-room model. Things stream in constantly; she processes them and calls out what is worth it.

Decision

- 1 **Observations never silently become facts about him.**
- 2 **She proposes, and he confirms.** "I saw a dentist appointment on your screen for Friday. Is that yours?" His yes writes it into me/ with his authority. His no, or no answer, leaves it an observation.
- 3 **Bridge questions are conversation openers.** Saliency decides which observations are worth raising and when, the same as any other call-out from the control room. They never interrupt; they wait for a natural opening or a quiet moment.
- 4 **Observations fade by retention policy; his facts do not.** A person forgets most of what they passively saw. Guests never agreed to being recorded forever, and camera and ambient audio fill a disk fast. Retention and saliency rules from Subcortical govern the observation store. Confirmed facts are kept until he changes them.

Rejected

- **Auto-promoting observations that look like facts about him.** That is the screen-overwrites-his-address failure, by design.
- **Never promoting observations.** It wastes the most useful thing an ambient observer can do: notice what he didn't think to tell her.

Consequences

Confirmation is his authority, applied: the same shape as ADR-0044 (a proposal, then his decision), and the same backstop as read-back (ADR-0057). Asking too often becomes nagging, so how often bridge questions are raised is itself a saliency setting, tuned by how he responds.